## Basic Info.

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| Model NO. | RAYTEK#G-9 |
| Warranty | 1 Year |
| Detection Method | Spectrophotometer |
| Advantage | High Resolution |
| Principle | New Spectrometer |
| Wavelengths | UV Spectrometer |
| Dispersive Element | Interferometer Spectrometer |
| Light Number | Monochromator |
| Certification | RoHS, ISO9001, CE |
| Customized | Non-Customized |
| Net Weight | 24kg |
| Wavelength Range | 190-1100nm |
| Spectral Bandwidth | 1.8nm/1nm(Optional) |

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| Wavelength Accuracy | ±0.1nm(656.1nm D2) |
| Transport Package | Wooden |
| Specification | 640*520*260 |
| Trademark | RAYTEK LASER |
| Origin | China Shanghai |
| HS Code | 8464901900 |
| Production Capacity | 1000piese/Year |

## Packaging & Delivery

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| Package Size | 70.00cm * 60.00cm * 35.00cm |
| Package Gross Weight | 50.000kg |

## Product Description

Product Description

The spectrophotometer G-9  is an optical analysis instrument released by Shanghai Ruitai, which integrates high precision and multiple functions. It is designed specifically for fields such as scientific research, teaching, and industrial testing. It adopts advanced laser technology and precise optical system, which can achieve fast and accurate spectrophotometric measurement, and is widely used in material composition analysis, structural identification, concentration determination and other fields.

Certifications

## Performance characteristics

•With the sample temperature control system, the sample temperature can be controlled between -5ºC and 200ºC.

•The lamp source calibration mechanism ensures that the instrument positions the lamp source optimally during each self-check.

•This feature eliminates the need for adjusting the light source after transportation due to vibration or after users replace the light source themselves.

• The optical system design and circuit control system design are extremely rigorous, and the selection and control of components are stringent, featuring high standards of accuracy, repeatability, low noise, and low stray light indicators.

•The open-style sample chamber can be equipped with reflective sample racks, automatic 5-way, 6-way, 8-way sample pool racks, solid sample racks, constant temperature water baths, and automatic samplers, suitable for various applications.

## Complete application functions

•Choosing broadband adjustable instruments can cater to different samples and testing requirements;

•Luminescence measurement: absorbance, transmittance. Multi wavelength fixed-point testing can simultaneously measure up to 30 wavelengths;

•Quantitative analysis: Automatically establish standard curves and store them for future testing and retrieval;

•Qualitative analysis: The scanning speed can be adjusted, the color filter position can be fine tuned, and the spectrum can be processed;

•Dynamics test: calculation of enzyme kinetics reaction rate;

•DNA/protein measurement: Built in specialized calculation formulas to directly obtain final results, making it easy to enter the field of life science research;

•Combined with the PC anti control workstation, the functions and data processing capabilities are more powerful.

## Application area

Spectrophotometers have a wide range of applications, let me briefly introduce the fields they are used in:

•Drug analysis: Pharmacopoeias in China and many countries around the world clearly stipulate that many drugs require the use of UV visible spectrophotometers for quality control. Therefore, UV visible spectrophotometers have become essential analytical instruments in the pharmaceutical and drug testing industries.

•Research institutions: National quality inspection agencies conduct quality inspections of products, quality control of manufacturing enterprises' finished products, inspection and quarantine of imported and exported goods, and research institutes.

•Teaching research: can be applied to the determination of complex composition, kinetic research, acid-base dissociation constant determination, and photometric titration.

•Environmental monitoring: Monitoring of water quality, atmosphere, rainfall, and soil can be completed according to national standards, and the content of various pollutants can be determined.

•Life Science: It can test trace samples in the field of life and provide DNA/protein detectors that can quickly measure the concentration of DNA/proteins.

•Manufacturing industry: can be applied to process control in industrial production, analysis of organic synthesis intermediates, analysis of additives, preservatives, and flavors in food, as well as determination of the content of nutrients such as fats, enzymes, sugars, minerals, vitamins, etc. in food, cosmetics, etc.

•Petrochemical industry: During the process of petroleum extraction and processing, petroleum may cause pollution, which can be applied to process control and finished product inspection in petrochemical production.

•Health and epidemic prevention: medical and health, clinical analysis, disease control analysis, trace sample testing in the field of life, analysis of human biochemical indicators, analysis of metabolites, drug analysis.

•Geological exploration: can be applied to the determination of metal elements and inorganic salts in minerals, marine geological exploration, hydrological stations, etc.

•Water quality analysis: can be applied to water treatment plants, sewage treatment plants, power plants, lake water quality testing stations, etc.

•Agriculture and forestry: can be applied to various specific industries in agriculture and forestry, such as pesticide residue detection, crop composition detection, fertilizer detection, soil composition detection, plant protection, plant quarantine, etc.

•Animal husbandry and fishery: can be applied to animal medicine testing, feed testing, animal disease prevention and control, etc.

•Metal analysis: detection of precious metal elements, detection of rare earth and alkaline earth metal elements, detection of harmful and beneficial elements

Product Parameters

| Type | G-9 | G-9S |
|---|---|---|
| Wavelength range | 190-1100nm | |
| Spectral bandwidth | 1.8nm/1nm(optional) | 0.5nm,1nm,2nm,4nm,5nm(optional) |
| Optical system | Dual beam, CT type optical path; 1200 line/millimeter laser holographic diffraction grating | |
| Wavelength accuracy | ±0.1nm(656.1nm D2),±0.3nm(all band) | |
| Wavelength repetition rate | ±0.1nm | |
| Wavelength Resolution | 0.01nm | |
| Photometric range | -4 to 4A;0-200%T;-9999 to 9999C | |
| Photometric accuracy | ±0.002A(0~0.5A);±0.004A(0.5~1.0A);±0.2%T(0~100%T) | |
| Photometric | ±0.001A(0~0.5A);±0.002A(0.5~1.0A);0.1%T(0~100%T) | |

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| Repeatability | | |
| Stray light | ≤0.03%T | |
| Baseline flatness | ±0.0008A(190~1100nm) | |
| Stability | ±0.0004A/hr @500nm,OA | |
| Photometric noise | ±0.0003A | |
| Data output | 320 × 240 icon display LCD screen<br>Connect USB to upper layer software | |
| Power supply | 80V~250V(power:120W) | |
| External dimensions | 640×520×260 | |
| Net weight | 24KG | 25KG |